

All sections should be as concise as possible and in narrative/paragraph form. You will need to balance between saying too little and saying too much. The former may seem like you are being concise but could make it hard for the reader to accurately follow your arguments, and the latter opens you up to more questions/critiques (“throwing spaghetti at the wall and seeing what sticks” is NOT good). The Results section tends to be the longest because of figures/tables/etc. that you will include in that section. Anything **bolded in the lab document** should find its way into the appropriate sections of the lab report. **Follow format guidelines in the course syllabus. You are expected to submit finished/polished reports, meaning everything in your report is proof-read, formatted properly, and understandable.** Check out papers published in research journals for more examples, such as the Journal of Biomedical Optics, Journal of Biophotonics Discovery, Scientific Reports, IEEE, etc.

- **Abstract (250 words max)**

- Broken up into the following sub-sections:
  - Significance (motivation for the work, with eye towards biomed instrumentation/applications)
  - Aim (this is the purpose of the presented work)
  - Approach (summary of methods)
  - Results (summary of key quantitative findings)
  - Conclusions (summary of major conclusions and implications for biomedical instruments/applications)
- Here’s an example from a journal paper: [LINK](#)

- **Introduction (max 1 page of text, not including figures/tables/equations/etc.)**

- Provide motivation for the lab that relates the focus/foci of the lab to biomedical instruments/instrumentation/signals/etc. For example, how/why the focus/foci of the lab are important in biomedical applications and why important to measure/understand/etc.
- Introduce and review relevant theory and concepts
- Include **in-line** citations where appropriate, see References section below for in-line formatting
- End this section with a paragraph that summarizes what you did in the lab

- **Methods**

- This section is about what you used and how you did it
- Should be in narrative/paragraph form (no bullet points or lists)
- Materials are usually embedded within the Methods descriptions with product number and vendor name in parentheses for each piece of equipment/software used
- You are encouraged to breakdown this section into sub-sections with appropriate header titles
- This section should include enough information and be written in such a way that someone of your level or higher should be able to reasonably replicate what you did from start to finish, including any analysis
- Figures/tables with captions and equations with equation numbers can go in this section to supplement your descriptions in this section, such as your own version of circuit diagrams/schematics, a diagram or photo of your set up with components CLEARLY labelled, LabVIEW programs with each part CLEARLY labeled, a table of known component values, etc.
- It should not read like a recipe. For example, don’t say “First we did ...”, “Then we did...”
- Do not direct the reader to the lab manual or say “...as described/instructed in the lab manual” or “See the lab manual for...” Do not state that you followed the procedure in the manual. Do not reference step numbers in the lab manual.
- Include **in-line** citations where appropriate, see References section below for in-line formatting

- **Results**

- Present results of collected and calculated data in plots and tables
- Figures/tables/etc. should be embedded in this section, NOT put in the appendix, and should have descriptive figure captions/table headings and should be numbered in the order they are presented

- Just report the results for what they are, present/state the facts, report values, etc., no interpretation or analysis of the results in this section (those go in the Discussion section), this is a purely objective reporting section
- Reference figure/table numbers in the actual text when referring to them, such as “In Figure 1,...”
- Back up qualitative statements (e.g., approximately, closely, etc.) with quantitative metrics (e.g., percent error, percent difference, etc.). Quantify your qualitative descriptions as much as possible, put numbers to qualitative statements, e.g., low/high or if say something lower/higher/smaller/greater, put some numbers to this comparison to make more quantitative
- Introduce statistics when appropriate (e.g., means, standard deviations, etc.): e.g., if claim something is significantly different from something else, did you compute a mean and standard deviation or standard error and run a statistical test? Additionally, you must report number of samples collected in this case. If you say a result is significant or some result is significantly greater/less than another result, then you MUST back that up with statistical tests for significance – fair warning though, make sure you are using statistical tests appropriately if you use them (e.g., making sure the data you’re performing the statistical test on abide by the assumptions the statistical test assumes about the data you’re assessing)
- Units on axes
- Make text for axis labels, tick mark labels and legends large – they should be able to be easily read without zooming into the document
- Don’t leave plots in default form, e.g., no default MATLAB plots, you must properly format and label them including axes titles, legends, etc.
- Must have text accompanying figures/tables/etc. We have seen before many figures/table/etc. without any text accompanying them or describing their results making it hard to follow along with the description of results in this section
- Always think to yourself, does your answer make sense? If not, you should go back and make sure your code is correct for solving for values or for plotting and then talk about potential sources of error in the Discussion section. This is a skill that will be instrumental in your future career as you may not know the expected result of an experiment so you need to be able to think through if your method makes sense and where it may have gone wrong

• **Discussion (max 2 pages of text, not including figures/tables/equations/etc.)**

- This is where you discuss what the results mean, your interpretation/analysis of the results, which includes answering questions from the lab document in narrative form. What do the results tell us? What went as expected? What was unexpected and what are some potential causes? What were the limitations of the work that provide context for the results presented? How can the experiment be improved if performed again? What would the next steps be if you were to continue the work?
- Having some discussion of the limitations of your results/work are important to include in this section
- Reference figure/table numbers in the actual text when discussing/analyzing/interpreting results, such as “As can be seen in Figure 1,...”
- Back up qualitative statements (e.g., approximately, closely, etc.) with quantitative metrics (e.g., percent error, percent difference, etc.). Quantify your qualitative descriptions as much as possible, put numbers to qualitative statements, e.g., low/high or if say something lower/higher/smaller/greater, put some numbers to these comparisons to make more quantitative
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- Statistics can be used to draw comparisons in a quantitative manner, summarize and compare grouped results (mean, median, mode, std dev, etc.)

- If you're going to posit/hypothesize (i.e., provide an explanation) for some observation of the data/results not rooted in something we've gone over in lecture, then you should find and cite appropriate references
- Include **in-line** citations where appropriate, see References section below for in-line formatting
- **Conclusion (5 sentences max)**
  - Major conclusions
  - Implications for biomedical instrumentation design and applications
- **References**
  - Cite references appropriately
  - This section contains a list of references in a numbered list, and the number assigned to the reference is the order in which you cite it in your lab report
  - Reference numbers should be **in-line** with your lab report text in brackets, e.g., [1] or [1-3] or [1,4-5] etc.
  - You're welcome to use any citation manager software
  - You may cite course material
  - Look at research papers from the Journal of Biomedical Optics, Journal of Biophotonics Discovery, Scientific Reports, IEEE, etc. for how to cite reference numbers in the main text and format listed references in this section
- **Appendix**
  - Including one is up to you, but **it will NOT be graded**, so make sure anything that should be graded is in the main lab report
- You can include an Acknowledgements section if you want to give credit to anyone who helped you with the lab in a way that warrants acknowledgement